

Joseph Walton

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Professional Summary

AI evaluation and machine learning engineer with over 10 years of experience building, testing and deploying production-grade AI, ML and software systems. Hands-on Python practitioner with experience establishing evaluation and observability practices, developing evaluation datasets and harnesses, red teaming Generative AI applications, and measuring model and application quality. Technical leader experienced in responsible AI, risk-aware delivery and communicating evaluation evidence to technical and non-technical stakeholders across government, international and commercial organisations.

AI Evaluation & Technical Skills

- **AI Evaluation & Assurance:** Evaluation strategy, automated evaluation harnesses, evaluation datasets, quality and relevance metrics, red teaming, adversarial testing, human-in-the-loop evaluation, observability, responsible AI
 - **LLM & Agentic Systems:** Azure OpenAI, Azure AI Foundry, Hugging Face, LangChain, LangGraph, retrieval-augmented generation (RAG), vector search
 - **Programming & Quality Engineering:** Python, SQL, Java, Git, Docker, Jupyter
 - **Machine Learning:** TensorFlow, PyTorch, Scikit-learn, MLflow, NLP, computer vision, recommendation systems
 - **Data Analysis:** Apache Spark, Pandas, Polars, NumPy, Matplotlib, Seaborn
 - **Cloud Platforms:** Azure, AWS, Google Cloud Platform
 - **Databases:** MySQL, PostgreSQL, Elasticsearch
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Professional Experience

IBM

Lead Data Scientist

June 2025 - Present

Contract role with central government.

- Rescued failing AI projects by introducing evaluation and observability best practices, improving teams' ability to assess quality and diagnose product behaviour.
- Defined evaluation approaches for agentic products and developed tooling to support repeatable, evidence-led assessment.
- Wrote a Claude Skill to optimise human-in-the-loop ingestion of cases into agent evaluation suites, reducing preparation time by approximately one hour per case.
- Designed and delivered a data processing agent using Azure AI Foundry, Azure Functions and LangGraph, saving two hours per case worker per day and approximately £2M annually.
- Rewrote a data matching algorithm in Polars from Apache Spark, reducing runtime from two hours to ten minutes and cloud costs by approximately £10k per month.

Tech Stack: Python, Azure AI Foundry, Azure OpenAI, LangGraph, Azure Functions, Polars

Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

Developed and evaluated NLP and Generative AI solutions for trade mark services within a public-sector environment.

- Developed a product roadmap for AI applications in the intellectual property domain, balancing user outcomes, technical feasibility and responsible delivery.
- Collaborated with multiple international intellectual property offices, communicating the UK's implementation of AI to technical and non-technical stakeholders.
- Contributed to cross-government AI policy and best practice as a member of a cross-government AI working group.
- Developed Generative AI applications within a service-oriented architecture using LangChain, Azure OpenAI and Azure Functions.
- Built a RAG agent to help novice users identify relevant Goods and Services terms, evaluating retrieval and application behaviour to guide refinement.
- Conducted red teaming and adversarial testing to evaluate and improve the robustness and safety of Generative AI applications.
- Implemented Elasticsearch vector search and tuned HNSW parameters to balance throughput and recall.
- Used quality and relevance metrics to drive iterative improvement of search and recommendation systems, increasing relevance by 20%.
- Implemented an end-to-end MLOps strategy with evaluation and release controls, reducing model deployment time to less than one day across multiple environments.
- Architected ETL pipelines using Polars, replacing Azure Databricks and reducing compute costs by 50% while enabling zero-downtime deployments.

Tech Stack: Python, Azure OpenAI, Hugging Face, LangChain, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

Led a team developing and assuring production ML systems for intellectual property protection.

- Led the design and implementation of an MLOps framework on Google Cloud Platform, using Vertex AI Pipelines to orchestrate training, evaluation, approval and deployment workflows.
- Established repeatable model evaluation and approval stages that reduced time to market to less than one day and supported client-specific deployments.
- Built Spark pipelines to prepare large-scale text and image datasets for training, evaluation, batch inference and recommendation workloads.
- Developed and deployed NLP and computer vision models, including a multilingual BERT model for detecting intellectual property infringements across more than 90 languages.
- Implemented a recommendation engine based on text and image similarity to improve detection of intellectual property violations.
- Mentored a software engineer through transition into a machine learning role.

Tech Stack: Python, Vertex AI Pipelines, TensorFlow, PySpark, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

Focused on the research, evaluation and production deployment of ML models for intellectual property protection.

- Engineered a production NLP system for predicting intellectual property infringement in a secure, containerised environment, increasing enforced infringements by 10% per analyst across approximately 200 analysts.
- Built and deployed an image similarity algorithm whose measured business performance helped secure a blue-chip client contract worth 10% of Annual Recurring Revenue.

Tech Stack: Python, TensorFlow, FastText, Google Cloud Platform, MLflow

Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

Contributed to national-scale data integration projects with a focus on secure data handling and public-sector delivery.

- Developed and evaluated ML models and data pipelines linking government administrative data in support of national surveys.
- Worked on the UK Census 2021 programme, developing a highly scalable backend for the Census data collection system.

Tech Stack: Python, PySpark, Cloudera

Barclays

Data Scientist

September 2018 - October 2018

Seconded to Barclays following success in the ONS x Barclays Hackathon.

- Developed and assessed regional economic models using payment data, with a focus on secure and ethical use of sensitive financial information.
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Purple Secure Systems

Software Engineer

August 2014 - October 2016

Developed full-stack big-data processing systems for clients in government and defence.

Tech Stack: Java, Python, PySpark, AngularJS

Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Honors & Awards

Winner of ONS x Barclays Data Hackathon

Used payments data to create more granular and timely estimates for Gross Value Added.